# MPM1D – Exam Review – Units 3-4 Homework

1. Gene works at a jean outlet store and she is paid $\$23/hr$. Let $h$ be the number of hours worked and let $E$ be her total earnings in dollars.

   a. Write an equation to represent Gene's earnings.

   b. Is this an example of direct variation or partial variation?

   c. What is the constant of variation?

   d. Construct a table of values to show Gene's earnings for $0, 1, 2, 3, 4, 5$ hours.

| | |
|---|---|
| | |
| | |
| | |
| | |
| | |
| | |

   e. How much will Gene earn if she works for 8 hours?

2. The volume of a barrel is $22\ litres$ but increases by $15\ litres$ per minute someone pours water into it. Let $V$ be total volume in litres and let $t$ be time in minutes.
   a. Write an equation to represent the volume of water in the barrel.

   b. Is this an example of direct variation or partial variation?

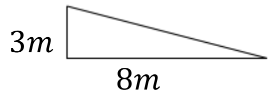
   c. What is the constant of variation?

   d. What is the constant or fixed amount?

   e. What is the volume of water in the barrel after someone pouring in water for 8 minutes?

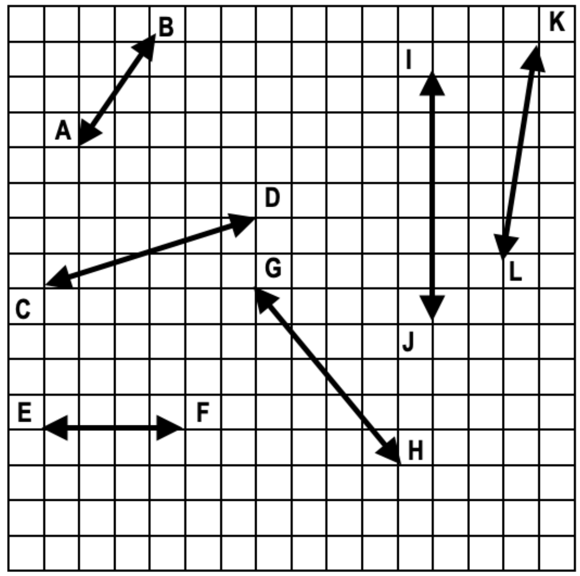
   f. How long will it take for the barrel to have a volume of $217$ litres?

3. Determine the slope of the ramp?




4. Determine the slope of each line. State the slope in lowest terms. Assume a one-unit grid.

| Line | Slope |
|---|---|
| AB | |
| CD | |
| EF | |
| GH | |
| IJ | |
| KL | |




5. Complete the table of values for the following linear relation and then write an equation to represent the linear relation.

| x | y |
|---|---|
| 0 | |
| 1 | 7 |
| 2 | 4 |
| 3 | 1 |
| | |
| | -5 |

6. A ramp rises $5\ m$ over a run of $15\ m$.
   a. How would you change the run to make the slope steeper?

   b. How would you change the rise to make the slope steeper?

7. A steel beam goes between the tops of two buildings. The first building is $32\ m$ high and the second building is $59\ m$ high. If the two buildings are $9\ m$ apart, what is the slope of the beam? Include a diagram in your solution.

8. Magic Mobile charges $\$37.50/hr$ to repair a cellphone in addition to an $\$85$ flat fee.
   a. Write an equation to represent the cost of getting a celelphone fixed at Magic Mobile. Let $C$ be the cost in dollars and let $h$ be the number of hours.

   b. Is this an example of direct variation or partial variation? Explain.

   c. How much would an individual have to pay if it took Magic Mobile three and one-half hours to repair their phone?

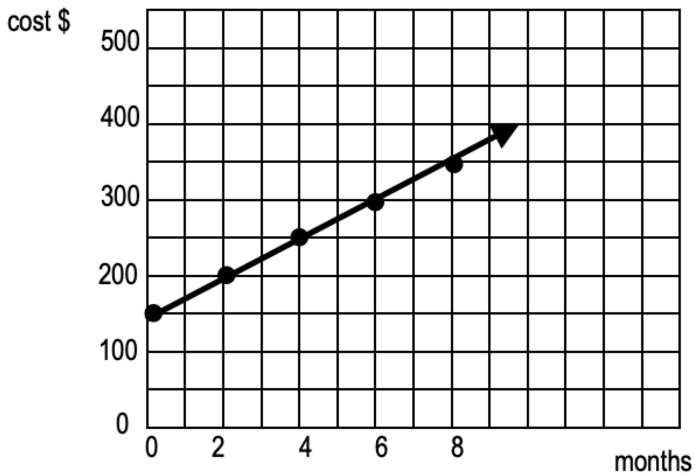
9. Mr. T and Mrs. T are renewing their wedding vows. The venue for the reception charges a fee of $\$12000$ to rent the venue and $\$175$ per person.

   a. Write an equation to represent the cost of renting the venue for this celebration. Let $C$ be cost in dollars and let $n$ be the number of people that attend.

   b. Use your equation to determine how much will it cost if $200$ people attend the reception.

   c. Use your equation to determine how many people attended the reception if the cost was $\$67475$.

10. The graph below represents the cost to join a gym.




a. Determine the slope of the line.

b. Explain the meaning of slope in this situation.

c. Determine the vertical intercept of the line.

d. Explain the meaning of the vertical intercept in this situation.

e. Write an equation to represent the cost of being a member of this gym.
   Define your variables.

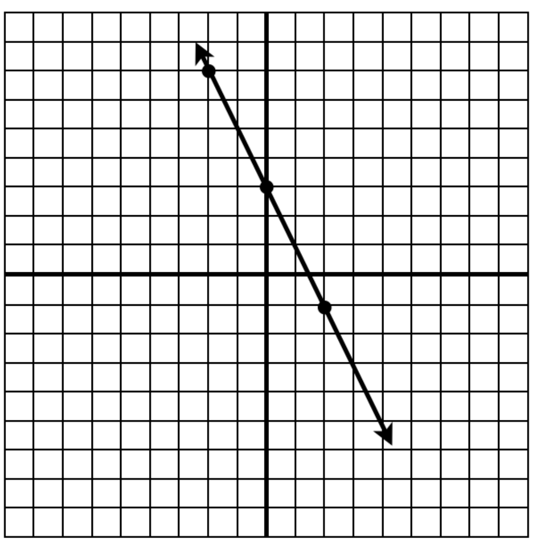
11. Using first differences, write an equation for each linear relation below.

| x | y |
|---|---|
| 0 | 20 |
| 1 | 21 |
| 2 | 23 |
| 3 | 25 |
| 4 | 31 |

| x | y |
|---|---|
| -3 | 27 |
| 0 | 31 |
| 3 | 35 |
| 6 | 39 |
| 9 | 43 |

| x | y |
|---|---|
| -3 | 0 |
| -2 | 5 |
| -1 | 10 |
| 0 | 15 |
| 1 | 20 |

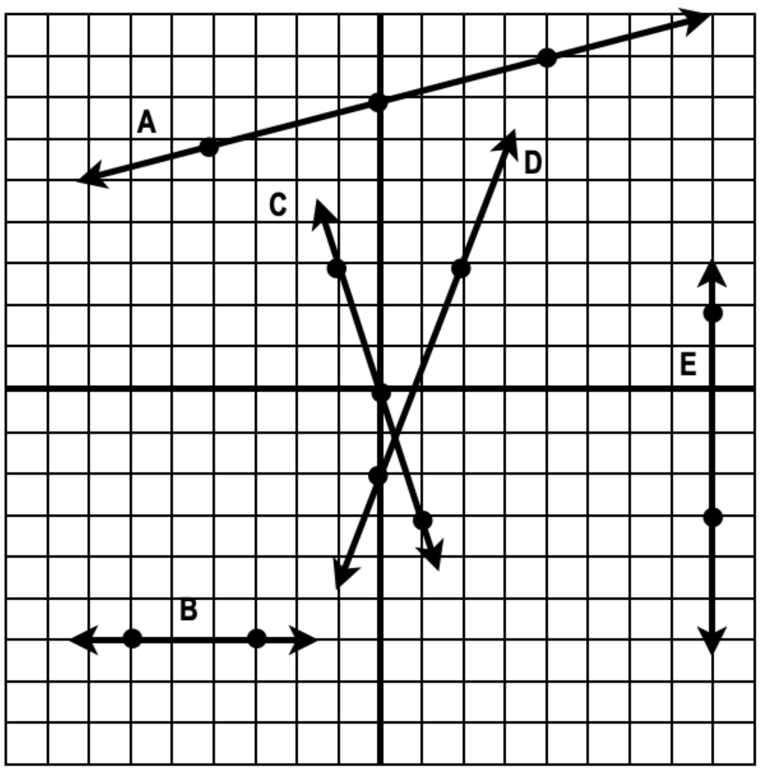
12. Use the RULE OF FOUR to represent the relation in three other ways. Assume the grid has a one-unit scale.



13. Identify the slope and y-intercept of each of the following linear equations.

| Equation | Slope | Y-Intercept |
|---|---|---|
| $y = 3x + 5$ | | |
| $y = -\frac{2}{3}x - 9$ | | |
| $y = 6$ | | |
| $x = -7$ | | |
| $3x - 6y + 12 = 0$ | | |

14. Write an equation in slope-intercept form for each of the following lines. Assume a one-unit grid.



15. Graph each of the following equations on the grid provided.

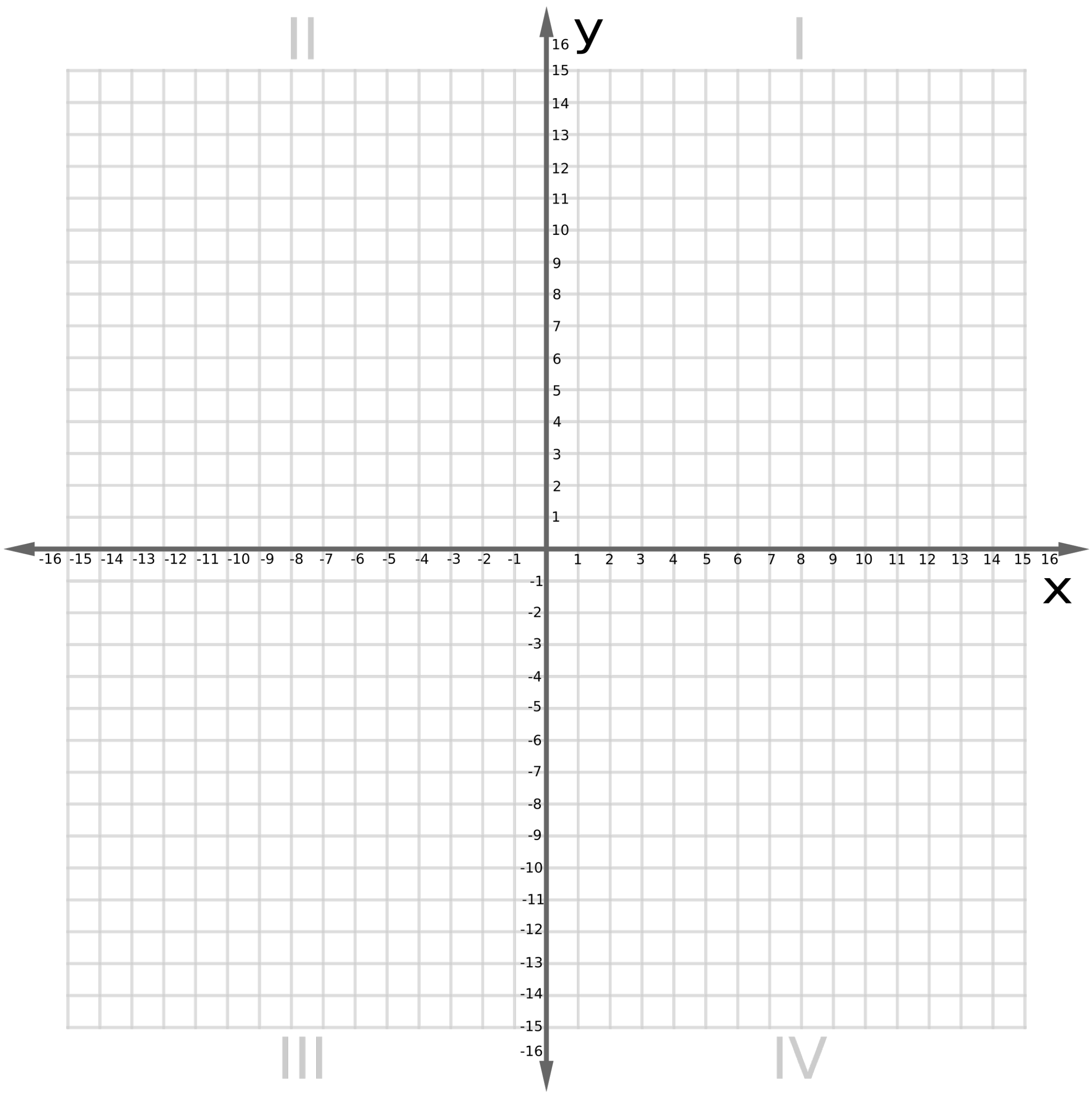
a. $y = x + 4$

b. $y = -\frac{3}{5}x - 2$

c. $2x + 4y - 12 = 0$

d. $y = -8$

e. $x = 7$



16. Rewrite each of the following equations. If the equation is in standard form, re-write it in slope-intercept form. If the equation is in slope-intercept form, re-write it in standard form.

a. $y = 3x - 8$

b. $y = -2x + 6$

c. $3x - 6y - 12 = 0$

d. $2x + 5y + 10 = 0$

17. Determine the slope of the line passing through $(-2, 5)$ and $(8, -1)$.

18. Determine the x-intercept and y-intercept of the line $4x - 3y - 24 = 0$. Write the coordinates of the intercepts.

19. Write a linear equation in slope-intercept form for each of the following lines:

a. A line passing through the point $(3, -7)$ with a slope of $2$.

b. A line passing through the points $(-3, 8)$ and $(-7, 16)$.

c. A line passing through $(-3, 8)$ with a slope perpendicular to the line $y = -\frac{1}{2}x + 9$.

d. A line parallel to $2x + 4y - 9 = 0$ and with the same y-intercept as $9x + 6y - 36 = 0$.

20. A blue spruce tree is planted and grows at a constant rate each year. After $9$ years, the tree is $458\ cm$ in height. After $23$ years the tree has a height of $976\ cm$.

    a. Determine the constant rate that the tree grows each year.

    b. Determine the height of the tree when it was planted.

    c. Write an equation to represent the height of the tree given the number of years it has been growing

    d. What will the height of the tree be after 37 years?

# Answer Key

1a) $E = 23h$ 1b) $Direct$ 1c) $23$ 1d) See Solutions 1e) $\$184$

2a) $V = 15t + 22$ 2b) $Partial$ 2c) $15$ 2d) $22$

2e) $142\ L$ 2f) 13 min

3) $m = \frac{3}{8}$

4) $AB: m = \frac{3}{2}$ $CD: m = \frac{1}{3}$ $EF: m = 0$ $GH: m = -\frac{5}{4}$ $IJ: m = undefined$ $KL: m = 6$

5) $y = -3x + 10$

6a) Decrease it 6b) Increase it

7) $m = 3$

8a) $C - 37.50h + 85$ 8b) Partial 8c) $\$216.25$

9) $C = 175n + 12000$ 9b) $\$47000$ 9c) 317 people

10a) $m = 25$ 10b) Gym membership costs $\$25$ every month

10c) $b = 150$ 10d) The gym charges a fixed cost of $\$150$ to join

10e) $C = 25m + 150$ Let $C$ be cost in \$ Let $m$ be the number of months

11a) Not Linear 11b) $y = \frac{4}{3}x + 31$ 11c) $y = 5x + 15$

12) $y = -2x + 3$ Y decreases by 2 when x increases by 1. Y starts at 3. Table: See solutions

13) $3{,}5$ 13b) $-\frac{2}{3}, -9$ 13c) $0, 6$

13d) undefined, none 13e) $\frac{1}{2}, 2$

14) $A: y = \frac{1}{4}x + 7$ $B: y = -6$ $C: y = -3x$ $D: y = \frac{5}{2}x - 2$ $E: x = 8$

15) See solutions

16a) $3x - y - 8 = 0$ 16b) $2x + y - 6 = 0$ 16c) $y = \frac{1}{2}x - 2$ 16d) $y = -\frac{2}{5}x - 2$

17) $m = -\frac{3}{5}$

18) $(6, 0) and\ (0, -8)$

19a) $y = 2x - 13$ 19b) $y = -2x + 2$ 19c) $y = 2x + 14$ 19d) $y = -\frac{1}{2}x + 6$

20a) $37\ cm/yr$ 20b) $125\ cm$

20c) $H = 37t + 125$ Let $H$ be height in cm. Let $t$ be time growing in years

20d) $1494\ cm$